A7.1 Single-copy atomicity size

The AXI4 protocol introduces the concept of single-copy atomicity size. This term defines the minimum number of bytes that a transaction updates atomically. The AXI4 protocol requires a transaction that is larger than the single-copy atomicity size to update memory in blocks of at least the single-copy atomicity size.

Note

Atomicity does not define the exact instant when the data is updated. What must be ensured is that no master can ever observe a partially updated form of the atomic data. For example, in many systems data structures such as linked lists are made up of 32-bit atomic elements. An atomic update of one of these elements requires that the entire 32-bit value is updated at the same time. It is not acceptable for any master to observe an update of only 16-bits at one time, and then the update of the other 16-bits at a later time.

More complex systems require support for larger atomic elements, in particular 64-bit atomic elements, so that masters can communicate using data structures that are based on these larger atomic elements.

The single-copy atomicity sizes that are supported in a system are important because all of the components involved in a given communication must support the required size of atomic element. If two masters are communicating through an interconnect and a single slave, then all of the components involved must ensure that transactions of the required size are treated atomically.

The AXI4 protocol does not require a specific single-copy atomicity size and systems can be designed to support different single-copy atomicity sizes.

Different groups of components can have different single-copy atomicity sizes for communication within the groups. In AXI4 the term single-copy atomic group describes a group of components that can communicate at a particular atomicity. For example, Figure A7-1 shows a system in which:

- the processor, Digital Signal Processor (DSP), DRAM controller, DMA controller, peripherals, SRAM memory and associated interconnect, are in a 32-bit single-copy atomic group
- the processor, DSP, DRAM controller, and associated interconnect, are also in a 64-bit single-copy atomic group.

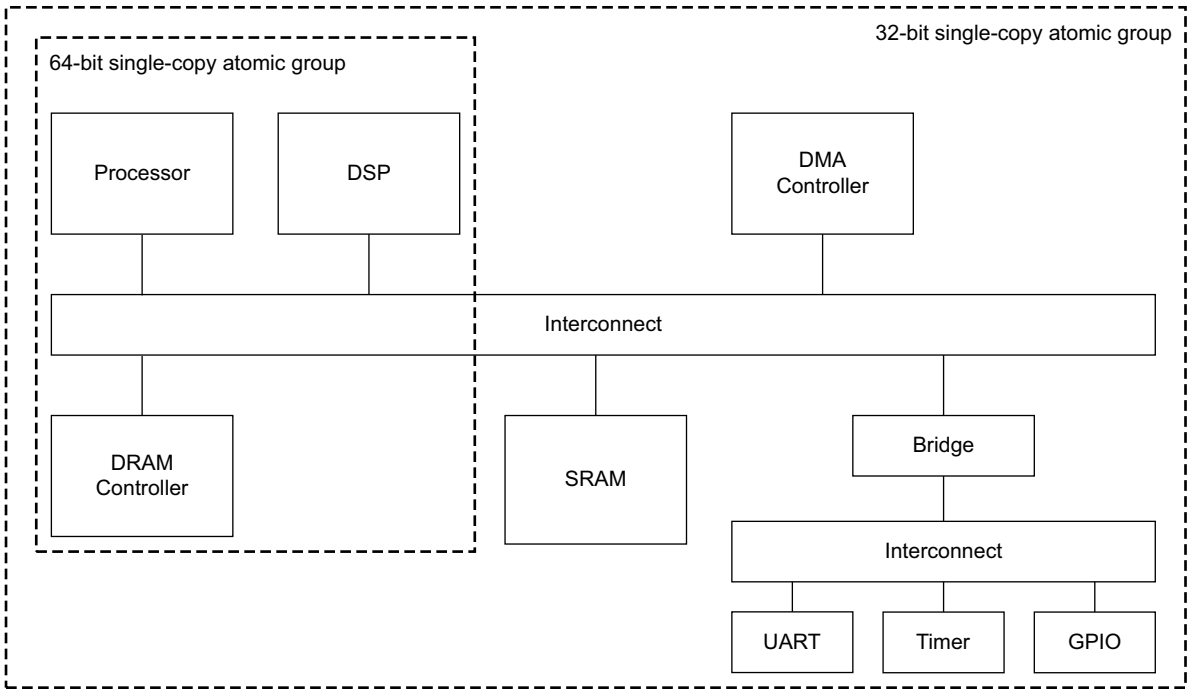


Figure A7-1 Example system with different single-copy atomic groups

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A transaction never has an atomicity guarantee greater than the alignment of its start address. For example, a burst in a 64-bit single-copy atomic group that is not aligned to an 8-byte boundary does not have any 64-bit single-copy atomic guarantee.

Byte strobes associated with a transaction do not affect the single-copy atomicity size.

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